

Synthesis of Ar-BINMOL Ligands by [1,2]-Wittig Rearrangement to Probe Their Catalytic Activity in 1,2-Addition Reactions of Aldehydes with Grignard Reagents

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We have demonstrated a highly diastereoselective synthesis of optically pure Ar-BINMOL-derived diols and their analogues. The present study demonstrates a unique cascade chirality transfer in a [1,2]-Wittig rearrangement that leads to chiral diols with three stereogenic centers, which include a chiral sp^3 center at the alcohol and C_2 -axial chirality. Screening these ligands in the arylation of aromatic aldehydes with Grignard reagents shows that the naphthyl-substituted BINMOL promotes the aryl transfer reaction in good yields (70–92 %) and moderate-to-good enantioselectivities

(up to 72 % ee), and a series of control experiments substantiates that the axial chirality and the chiral sp^3 center at the alcohol of the Ar-BINMOLs are the pivotal enantioselectivity-controlling structure elements. In addition, this study demonstrated the importance of the chiral sp^3 center at the alcohol on Ar-BINMOL for the aryl transfer reaction. Finally, we found that the chiral Ar-BINMOL ligand **2h** mediated the titanium-promoted 1,2-addition of MeMgBr to aldehydes to give the desired products in good yields with excellent enantioselectivities (up to 92 % ee).

Introduction

Chiral arylmethanols are key structural elements of many biologically active compounds and optoelectronic molecules, which have found wide application as agrochemicals and pharmaceuticals.^[1] Therefore, the synthesis of chiral arylmethanols has attracted much attention in the past decades. However, the enantioselective synthesis of diarylmethanols is not an easy task. Among the different approaches to diarylmethanols, there are two major protocols for their enantioselective syntheses: the asymmetric reduction of unsymmetric diaryl ketones^[2] and the enantioselective addition of an aryl–metal reagent to aromatic aldehydes.^[3] Aryl transfer to aldehydes represents a powerful and widely used transformation and is a straightforward route to enantiomerically enriched diarylmethanols.^[4] Since Seebach and Weber reported the first catalytic phenyl transfer reaction of PhLi,^[5] several groups have discovered enantioselective catalysts based on amino alcohols and diols for

the addition of an aryl–metal reagent to aromatic aldehydes.^[3,6] However, most enantioselective arylations of aldehydes rely on the use of expensive diphenylzinc reagents.^[7] A more practical and versatile method would begin with an aryl bromide or the corresponding Grignard reagent, many of which are commercially available and inexpensive. Although the addition of Grignard reagents to aldehydes is an ideal and easy method to prepare the racemic secondary alcohols, catalytic enantioselective Grignard reactions with aldehydes have rarely been reported because of the high and uncontrollable reactivity of Grignard reagents with aromatic aldehydes.^[8] It is well known that the lack of suitable catalyst systems has limited their further development during recent decades.^[9] Recently, Harada et al.^[10] reported an efficient method for the asymmetric Grignard addition of aryl reagents to aldehydes in high enantioselectivities, which was a significant improvement upon previous work. The key to the success of this reaction was the use of 3-(3,5-diphenylphenyl)-BINOL ligand (DPP-BINOL). For the investigation of a new catalyst system, the design and synthesis of ligands is a key challenge to achieve enantioselective arylation reaction of aldehydes.

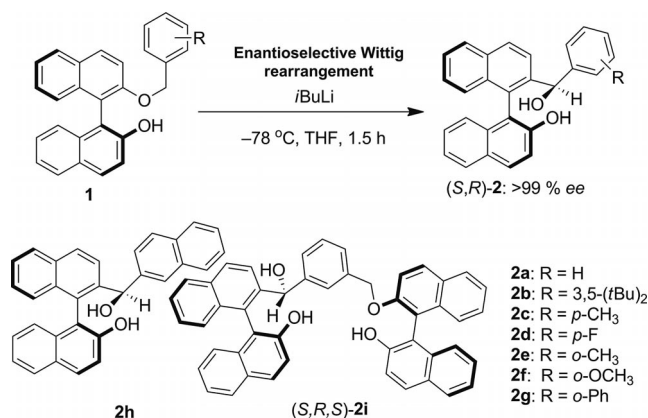
Very recently, we have developed a facile and practical methodology for the preparation of new chiral diols, in which the simple axial chiral monoalkylated BINOLs could be converted into synthetically useful and enantiomerically pure 1,1'-binaphthalene-2- α -arylmethanol-2'-ols (Ar-BINMOLs) with axial and sp^3 -central chirality through the axial-to-central chirality transfer of a [1,2]-Wittig re-

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arrangement.^[11] In this method, the chirality transfer process showed very broad substrate scope in terms of additional aromatic ethers with excellent enantioselectivities and yields, and thus allows access to a broad range of chiral Ar-BINMOLs. Over the past year, it has proved to be an attractive and promising chiral ligand in several transformations.^[12] Herein, we report the preparation of several new chiral diol ligands endowed with axial and sp^3 -central chirality and their application to the titanium-promoted asymmetric 1,2-addition of aryl and methyl Grignard reagents to aryl aldehydes to give diarylmethanols with good enantioselectivities. More importantly, we report a new strategy to prepare some Ar-BINMOL ligands by a double [1,2]-Wittig rearrangement, which provides versatile, optically pure compounds for organic synthesis (Scheme 1).



Scheme 1. Synthesis of Ar-BINMOLs by an asymmetric [1,2]-Wittig rearrangement.^[11,12a]

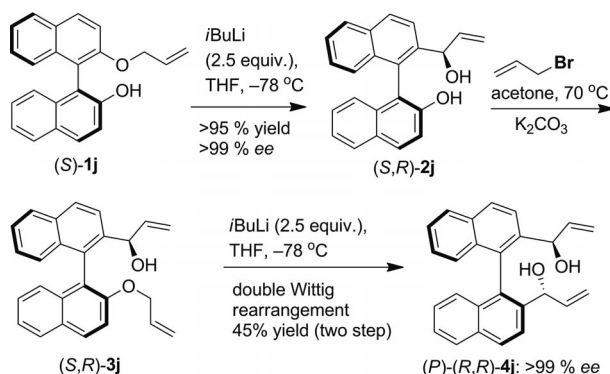
Results and Discussion

Chiral diols represent a very useful class of chiral ligands that have been widely used as optically pure catalysts and chiral auxiliaries; therefore, much effort has been dedicated to practical procedures for their synthesis. Although the [1,2]-Wittig rearrangement is recognized as a powerful method for the synthesis of secondary alcohols by carbon-oxygen cleavage and carbon-carbon bond forming reactions and its mechanism and stereochemistry is well studied, its utility in organic synthesis has been limited by rather low yields, restricted substrate range, and the lack of high diastereo- or stereoselectivity.^[13] To prepare new chiral BINOL-derived diol ligands, we have developed a strategy with neighboring lithium activation for the asymmetric [1,2]-Wittig rearrangement. This [1,2]-Wittig rearrangement process showed very broad substrate scope with excellent enantioselectivities (> 99% *ee*) and yields (84–96%). In a further study of the application of Ar-BINMOLs in asymmetric catalysis, we have found that the Ar-BINMOLs could successfully act as ligands to promote the asymmetric 1,2-addition of diethylzinc to aldehydes and resulted in excellent yields and enantioselectivities (up to > 99.9% *ee*) in the reactions of diethylzinc with a broad range of aromatic

aldehydes.^[12a] These observations prompted us to investigate further syntheses of new Ar-BINMOLs and other diols by [1,2]-Wittig rearrangements and to explore the possibility of enantioselective arylation of aromatic aldehydes with Grignard reagents.

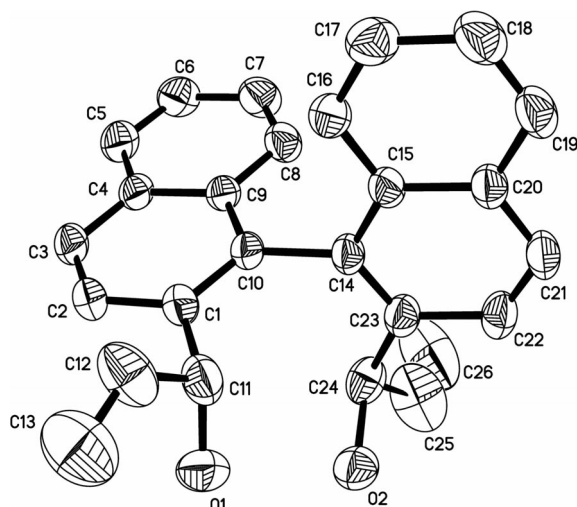
Synthesis of New Ar-BINMOLs and Diol Ligands by [1,2]-Wittig Rearrangement

Efforts to design the diol ligands focused on structural modifications that retained the sp^3 -central chirality of the alcohol and the C_2 -axial chirality and were performed with [1,2]-Wittig rearrangements. Therefore, we designed and synthesized new chiral diols from BINOL (Scheme 2). As benzyl ethers can lead to efficient [1,2]-Wittig rearrangements, we turned our attention to the asymmetric [1,2]-Wittig rearrangement of a BINOL-containing allylic ether. As shown in Scheme 2, the rearrangement of the protected mono-allylic ether **1j** occurred in the presence of isobutyl-lithium and yielded the corresponding chiral diol **2j** in excellent yield with perfect chirality transfer and enantioselectivity (> 99% *ee*). With the diol **2j** in hand, we examined the cascade [1,2]-Wittig rearrangement to produce the optically pure diol **4j**. The absolute stereochemistry of **4j** was determined by X-ray analysis. The crystal structure of (*P*)-(*R,R*)-**4j** (Figure 1) shows that the highly diastereoselective cascade chirality transfer through a [1,2]-Wittig rearrangement takes place with the aid of neighboring lithium activation.

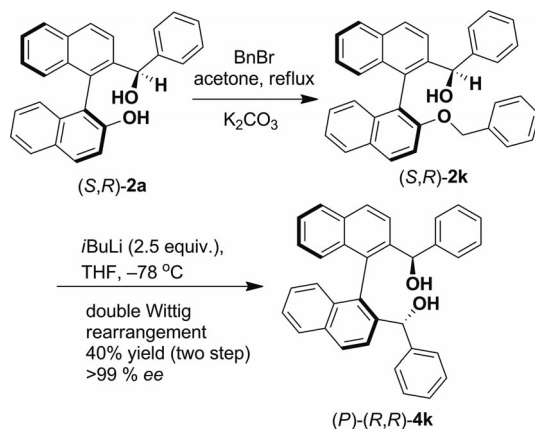


Scheme 2. Synthesis of chiral allylic BINMOL derivative **4j** by cascade chirality transfer in a [1,2]-Wittig rearrangement.

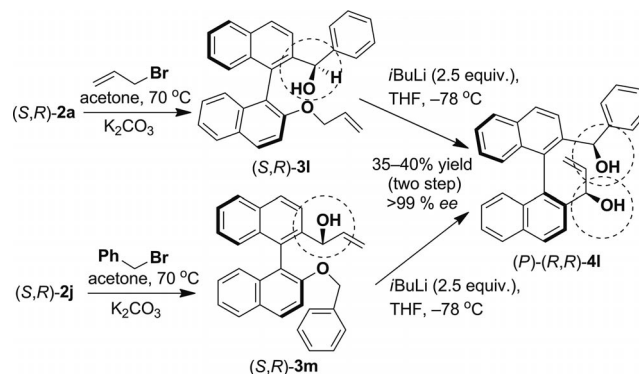
To examine the cascade chirality transfer in the [1,2]-Wittig rearrangement of BINMOL analogues, we synthesized the benzyl Ph-BINMOL **2k**, in which the naphthol moiety is protected but the alcohol is unbenzylated (Scheme 3). In the next cascade rearrangement, we found the desired product was obtained with perfect chirality transfer to give only one diastereomer (Scheme 3). Although the yield is not ideal, it is a practical method for the preparation of diphenyl-substituted chiral secondary alcohols with axial and sp^3 -central chirality in excellent enantioselectivity (> 99%

Figure 1. The crystal structure of **4j**.

ee). In analogy to a previous study on the axial-to-central chirality transfer of [1,2]-Wittig rearrangement and to further study the effect of a neighboring stereogenic alcohol, we examined the cascade [1,2]-Wittig rearrangement of a structurally divergent ether. Interestingly, the allyl- or benzyl-substituted BINMOL (**3l** or **3m**) led to the same chiral product (*P*)-(*R,R*)-**4l** (Scheme 4). On the basis of this finding, the absolute configuration of **4l** from (*S,R*)-**3l** bearing (*R*)-phenylmethanol and (*S,R*)-**3m** bearing (*R*)-allylic alcohol was determined to be the (*S,R,R*) or (*P*)-(*R,R*) conformer. This data also further reinforces the conclusion that neighboring lithium activation is beneficial to the [1,2]-Wittig rearrangement.

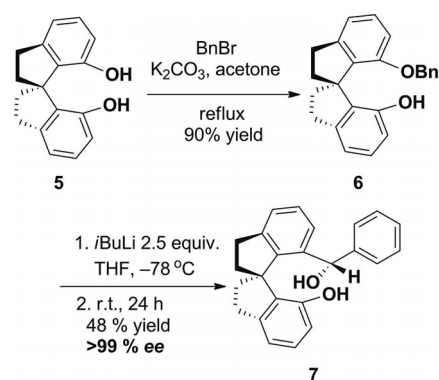
Scheme 3. Synthesis of chiral allylic BINMOL derivative **4k** by cascade chirality transfer in a [1,2]-Wittig rearrangement.

These promising results encouraged us to extend the scope of the [1,2]-Wittig rearrangement to 1,1'-spirobiindan-7,7'-diol (**5**). 1,1'-Spirobiindan-7,7'-diol belongs to the family of C_2 -symmetric ligands. The structure of 1,1'-spirobiindan offers a promising combination of chemical robustness and conformational rigidity. In the past decade,

Scheme 4. Synthesis of chiral allylic BINMOL derivative **4l** by cascade chirality transfer in a [1,2]-Wittig rearrangement.

a large number of 1,1'-spirobiindan derivatives, including chiral 1,1'-spirobiindan-7,7'-diol (SPINOL) and its P- or N-containing spiro chiral ligands,^[14] have been prepared and their potential in asymmetric catalysis has been revealed.^[15] As an alternative to BINOL derivatives and Ar-BINMOLs, we assume that 1,1'-spirobiindan-7,7'-diol can be effectively modified by introducing a bulky phenyl group and stereogenic secondary alcohol carbon-center to form a new diol with axial and sp^3 -central chirality by axial-to-central chirality transfer based on diastereoselective neighboring lithium-assisted [1,2]-Wittig rearrangement.

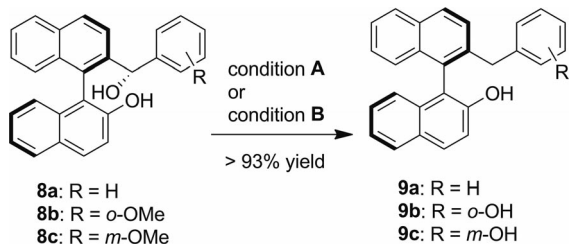
As demonstrated in Scheme 5, the preparation of diol **7** (Ph-SPINMOL) commenced with **5**. Benzylation of **5** followed by a butyllithium-promoted [1,2]-Wittig rearrangement at room temperature gave the desired diol **7** in moderate yield with excellent chirality transfer (CT). As expected and similarly to the axial-to-central chirality transfer of the [1,2]-Wittig rearrangement of chiral monoalkylated BINOLs, the Ph-SPINMOL was confirmed as an optically pure chiral molecule (> 99% *ee*, see Supporting Information).



Scheme 5. The synthesis of a new SPINOL-derived diol (Ph-SPINMOL) by [1,2]-Wittig rearrangement.

To illustrate the synthetic potential and structural characteristics of these chiral Ar-BINMOLs and analogous compounds in asymmetric catalysis, we also prepared some diols without chiral sp^3 carbon atoms derived from Ar-

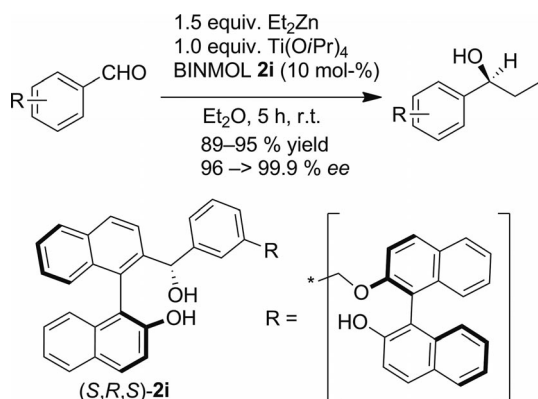
BINMOLs (Scheme 6) and used them to check the crucial role of the alcoholic sp^3 chiral center in subsequent arylations of aromatic aldehydes with Grignard reagents. The Ar-BINMOLs **8a–c** were treated with trimethylsilyl chloride (TMSCl)/NaI^[16] to give the corresponding deoxygenated products **9a–c** in high yields.



Scheme 6. Synthesis of deoxygenated phenol ligands **9a–c**. Condition A for **8a**: TMSCl/NaI (6 equiv.) in CH_3CN , room temperature. Condition B for **8b–c**: TMSCl/NaI (10 equiv.) in CH_3CN , reflux.

Ar-BINMOL/Chiral Diol Ligands in the Titanium-Promoted 1,2-Addition of Aldehydes with Grignard Reagents

With the Ar-BINMOLs and analogous compounds in hand, we immediately sought to establish the catalytic potential of these diols in asymmetric reactions. In our previous report, we have shown that excellent results were achieved with Ar-BINMOLs in the asymmetric 1,2-addition of diethylzinc to aldehydes (Scheme 7).^[12] Therefore, we had a great interest to test our ligands for the arylation of aldehydes with Grignard reagents.

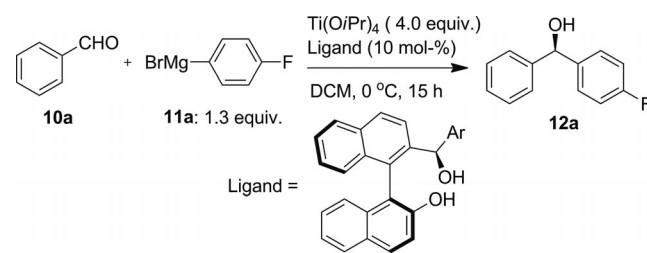


Scheme 7. Enantioselective addition of diethylzinc to aldehydes using BINMOL **2i**.

Our study began with a preliminary screening of BINOL and several of the Ar-BINMOLs shown in Scheme 1. By using benzaldehyde (**10a**), and (4-fluorophenyl)magnesium bromide (**11a**) as a probing substrate, the reaction was performed with $Ti(OiPr)_4$ at 0 °C, and the results are summarized in Table 1. Although both tetrahydrofuran (THF) and diethyl ether (Et_2O) were good solvents for the arylation of

aldehydes with Grignard reagents and gave good yields in the presence of Ph-BINMOL **2a**, the enantioselectivities were low (35–39% *ee*). The solvent effects were evaluated with the ligand **2a** and dichloromethane (DCM) was found to be the best solvent that gave promising results in term of enantioselectivity (46% *ee*). Ar-BINMOLs with either electron-withdrawing or electron-donating substituents (**2b–2f**) as well as a phenyl-substituted chiral diol (**2g**) were applied to promote the arylation of aldehyde. When using chiral Ar-BINMOLs **2b–d**, the desired product **12a** was obtained in 36–56% *ee*. Interestingly, *o*-methyl-substituted **2e** and naphthyl-substituted **2h** gave the same level of enantioselectivity (65 or 66% *ee*) in this reaction, and the use of BINOL or BINOL-linked Ar-BINMOL **2i** resulted in decreased enantioselectivities (38 and 57% *ee*, respectively). However, there were almost no difference in the enantioselectivity when either **2e** or **2h** was used. The slight difference in the catalytic activity of **2e** and **2h** in the model 1,2-addition reaction of benzaldehyde and (4-fluorophenyl)magnesium bromide propelled us to conduct further ligand screening in the model arylation of 2-methoxybenzaldehyde (**10b**) with phenylmagnesium bromide (**11b**).

Table 1. Preliminary screening of chiral Ar-BINMOL ligands for the asymmetric 1,2-addition of (4-fluorophenyl)magnesium bromide to benzaldehyde.^[a]



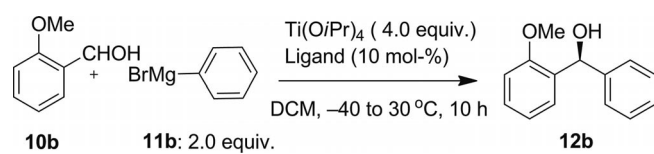
Entry	Ligand	Solvent	Yield [%] ^[b]	<i>ee</i> [%] ^[c]
1	2a	THF	64	35
2	2a	Et_2O	74	39
3	2a	DCM	54	46
4	2b	DCM	59	47
5	2c	DCM	54	56
6	2d	DCM	59	49
7	2e	DCM	54	66
8	2h	DCM	64	65
9	2i	DCM	59	57
10	(<i>S</i>)-BINOL	DCM	50	38

[a] 1 equiv. of Grignard reagent was treated with 2 equiv. of $Ti(OiPr)_4$ at 0 °C for 1–2 h, and then the mixture of Grignard reagent and titanium was added to the solution of benzaldehyde (1 equiv.), Ar-BINMOL or BINOL ligand (10 mol-%), and $Ti(OiPr)_4$ (2 equiv.). The reaction was carried out at 0 °C for 15 h. [b] Isolated yields. [c] Determined by HPLC with a Daicel CHIRALCEL OB-H column.

Further optimization studies with chiral Ar-BINMOLs **2a–2i** and various Ar-BINMOL-derived ligands (**4–9**) under different conditions are summarized in Table 2. Compounds **2e** and **2h** exhibited promising enantioselectivities (65–66% *ee*) in the reaction of benzaldehyde (**10a**) and (4-

fluorophenyl)magnesium bromide (**11a**), and we found that **2h** is superior to **2e** in the arylation of 2-methoxybenzaldehyde (**10b**) with phenylmagnesium bromide (**11b**). Therefore, we continued to examine the enantioselective induction of new Ar-BINMOLs (**2f–j**) and their derivatives (**4** and **9**) as well as SPINOL-derived diol **7** as ligands in this reaction. These ligands were screened for the arylation under modified conditions (Table 2, Entries 3–15). The highest efficiency and enantioselectivity for the reaction was still observed with Ar-BINMOL **2h**. Interestingly, although **2f** resulted in low enantioselectivity (36% *ee*), the *o*-methoxy group on this ligand changed the stereochemical outcome

Table 2. Optimization of reaction conditions with chiral Ar-BINMOL ligands.^[a]



Entry	Ligand	Solvent	Temp. [°C]	Yield [%] ^[b]	<i>ee</i> [%] ^[c]
1	2h	DCM	–20	71	71
2	2e	DCM	–20	72	63
3	2f	DCM	–20	70	–36
4	2g ^[d]	DCM	–20	76	–51
5	2i	DCM	–20	77	61
6	2j ^[d]	DCM	–20	75	–2
7	4j	DCM	–20	82	3
8	4k	DCM	–20	79	6
9	4l	DCM	–20	77	4
10	5	DCM	–20	72	–21
11	7	DCM	–20	87	0
12	9a	DCM	–20	35	0
14	9b	DCM	–20	87	–43
15	9c	DCM	–20	86	0
16	2h	Et ₂ O	–20	81	41
17	2h	toluene	–20	98	55
18	2h	DCM	30	55	42
19	2h	DCM	0	65	63
20	2h	DCM	–10	65	65
21	2h	DCM	–40	47	61

[a] 1 equiv. of Grignard reagent was treated with 2 equiv. of Ti(OiPr)₄ at 0 °C for 1–2 h, and then the mixture of Grignard reagent and titanium was added to the solution of benzaldehyde (1 equiv.), ligand (10 mol-%), and Ti(OiPr)₄ (2 equiv.). The reaction was carried out at –40 to 30 °C for 10 h. [b] Isolated yields. [c] Determined by HPLC with a Daicel CHIRALCEL OJ-H column. [d] (*R,S*) conformer derived from (*R*)-BINOL.

and the absolute configuration is opposite to that of **2e** or **2h**, which can clearly be explained by the cooperative effect of the chiral sp³ center of the alcohol and the neighboring MeO group on the titanium complex. Moreover, the introduction of a phenyl ring at the *ortho*-position of Ph-BINMOL (for example, **2g**) did not improve the enantioselectivity in comparison to that of *o*-methyl-substituted Ar-BINMOL **2e**. When using the allylic BINMOL **2j** as ligand, almost no enantioselectivity was observed in this reaction, which confirmed the importance of the CH(OH)-linked aromatic ring on the Ar-BINMOL ligands.

Unfortunately, with **4j–l** prepared from cascade diastereoselective [1,2]-Wittig rearrangement as ligand, the enantioselectivity decreased to only 3–6% *ee* (Table 2, Entries 7 and 8). In addition, although chiral 1,1'-spirobiindan-7,7'-diol (SPINOL) could also promote this reaction smoothly, the enantioselectivity was rather poor (Table 2, Entry 10). More interestingly, when using diol **7** prepared from SPINOL, the product **12b** obtained was racemic.

To support the hypothesis that a chiral sp³ alcohol or secondary alcohol moiety (CH–OH) is crucial to the titanium-promoted arylation of aldehyde (Figure 2), we evaluated the catalytic activity of **9** with or without a phenolic group (Table 2, Entries 12–14). The beneficial effects of a secondary alcohol moiety on Ar-BINMOLs were confirmed in this reaction, in which both **9a** without an additional alcohol or phenol moiety and **9c** with a long-distance *meso*-position phenol moiety resulted in racemic product. These illustrations clearly demonstrate that the Ar-BINMOL analogues without an alcoholic moiety (CH–OH) lead to the formation of inactive titanium catalysts (Figure 2, **II** and **III**)^[17] in enantioselective induction. Further evidence is that **9b** with a capable neighboring phenol gives **12b** in comparable enantioselectivity (43% *ee*).

The solvent effects were also evaluated with the ligand **2h** for the arylation of 2-methoxybenzaldehyde (**10b**) with phenylmagnesium bromide (**11b**). These results continued to confirm that DCM was the best solvent in terms of enantioselectivity (Table 2, Entries 16 and 17). Efforts to improve the enantioselectivity by adjusting the reaction temperature (Table 2, Entries 18–21) and the amount of Ti(OiPr)₄ were unsuccessful.

Based on the experimental validation of the stereocontrolling ability of the Ar-BINMOL **2h**, we further examined

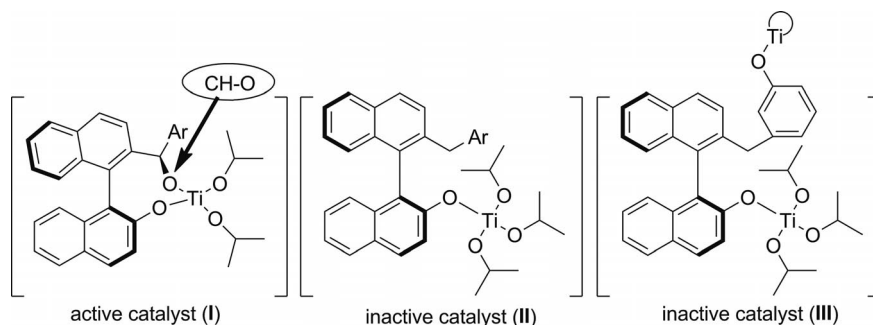


Figure 2. Possible titanium complex derived from Ar-BINMOL – the importance of the alcohol (CH–OH).

the generality of the present arylation of aromatic aldehydes with Grignard reagents and representative results are summarized in Table 3. With several Grignard reagents in hand, a wide range of aromatic aldehydes with different substituents on aromatic rings, including methyl, methoxy, and halide substituents, could be used as electrophiles, and good to excellent yields were uniformly obtained. As shown in Table 3, although moderate-to-good enantioselectivities were obtained in the arylation of aldehydes, variation of the electronic and steric nature of the Grignard reagents or aromatic aldehyde appeared to have little influence on the asymmetric induction in the presence of ligand **2h**. These results clearly demonstrate that Ar-BINMOL **2h** as a chiral ligand can give promising and acceptable enantioselectivities in the direct arylation of aldehydes with Grignard reagents. In addition, it shows that the development of new chiral diols or analogues by modification of the backbone of chiral Ar-BINMOLs would be attractive.

Table 3. Synthesis of chiral diarylmethanols through arylation of aldehydes with Grignard reagents.^[a]

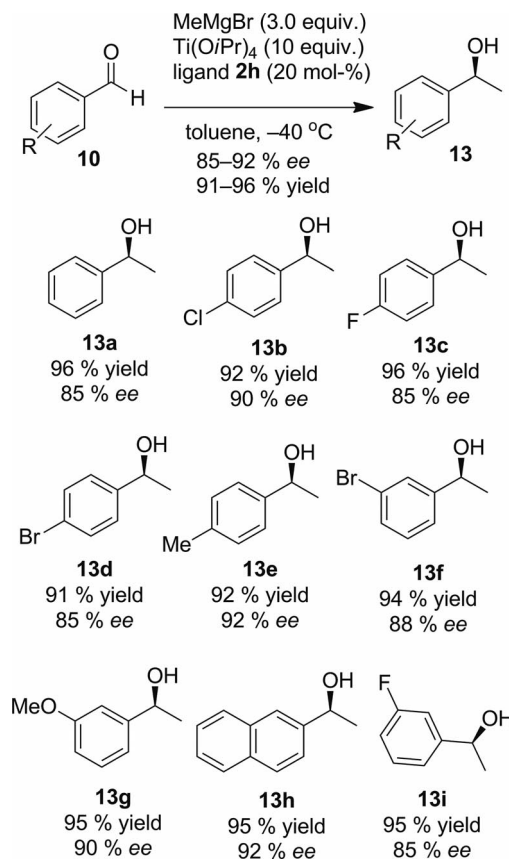


Entry	R	R'	Yield [%] ^[b]	ee [%] ^[c]
1	<i>o</i> -CH ₃	H	12b : 84	70
2	<i>m</i> -OCH ₃	H	12d : 91	55
3	<i>m</i> -F	H	12e : 91	54
4	<i>m</i> -Br	H	12f : 92	50
5	<i>p</i> -CH ₃	H	12g : 85	50
6	<i>p</i> -Br	H	12h : 76	55
7	<i>p</i> -OCH ₃	H	12i : 70	61
8	naphthyl	H	12j : 77	54
9	<i>o</i> -CH ₃	F	12k : 74	65
10	<i>o</i> -OCH ₃	F	12l : 83	69
11	<i>m</i> -Br	F	12m : 76	56
12	<i>m</i> -F	F	12n : 82	54
13	<i>o</i> -CH ₃	<i>p</i> -MeO	12o : 77	72
14	<i>o</i> -Br	<i>p</i> -MeO	12p : 85	55
15	<i>o</i> -Cl	<i>p</i> -MeO	12q : 72	66
16	H	<i>p</i> -MeO	12r : 89	69
17	<i>m</i> -Br	<i>p</i> -MeO	12s : 70	71

[a] In a typical procedure, 1 equiv. of Grignard reagent was treated with 2 equiv. of Ti(OiPr)₄ at -78 °C for 1–2 h, and then the mixture of Grignard reagent (1.0 equiv.) and titanium was added to the solution of aromatic aldehyde (1 equiv.), Ar-BINMOL ligand **2h** (10 mol-%), and Ti(OiPr)₄ (2 equiv.). The reaction was carried out at -20 °C for 10 h. [b] Isolated yields. [c] Determined by HPLC.

On the basis of the above findings, further studies on the synthetic application of Ar-BINMOL **2h** in the asymmetric 1,2-addition of MeMgBr to aldehydes were carried out under the modified conditions similar to those of a previous report.^[12b] The results from scope studies for the enantioselective 1,2-addition of MeMgBr to aldehydes with Ar-BINMOL ligand **2h** are shown in Scheme 8. Reactions with substituted aromatic aldehydes were highly effective in conversions and enantioselectivities. Halide as well as methyl substituents on the aromatic ring of the aldehydes are also well tolerated in this reaction, which leads to the chiral aryl-

methanols **13a–i** with good enantioselectivities (85–92% *ee*). Therefore, an enantioselectivity enhancement with **2h** in comparison to the previous method^[12b] is observed in the reaction, in which unprecedented yields and enantioselectivities are achieved in the asymmetric methylation of aldehydes.



Scheme 8. Asymmetric 1,2-addition of MeMgBr to aromatic aldehydes.

Conclusions

We have demonstrated a highly diastereoselective synthesis of optically pure Ar-BINMOL-derived diols and their analogues by an enantioselective [1,2]-Wittig rearrangement with the aid of the directing effect of functional group/ neighboring lithium atom. The present study demonstrates unique cascade or relay chirality transfer in a [1,2]-Wittig rearrangement that leads to chiral diols with three stereogenic centers, including a chiral secondary alcohol and C₂-axial chirality. In addition, we report the first synthesis of a SPINOL-derived diol (Ph-SPINMOL) prepared from chiral 1,1'-spirobiindan-7,7'-diol through a highly diastereoselective [1,2]-Wittig rearrangement, which further disclosed the potential of [1,2]-Wittig rearrangements for the enantioselective synthesis of chiral diols. To probe the enantioselective induction activity of these chiral diols prepared through [1,2]-Wittig rearrangements, the synthesis of chiral diarylmethanols by arylation of aromatic aldehydes

with Grignard reagents was selected as a model. The ligand screening results show that the naphthyl-substituted BINMOL **2h** promotes this aryl transfer reaction in good yields and moderate-to-good enantioselectivities, and a series of control experiments substantiate that both the axial and sp^3 central chirality of the Ar-BINMOLs are the pivotal enantioselectivity-controlling structure elements in the catalytic arylation of aldehydes with aryl Grignard reagents. In addition, this study demonstrated the importance of the sp^3 chiral center at the alcohol on Ar-BINMOL for the aryl transfer reaction. Although the arylation reaction of aldehydes with Grignard reagents is not a good model for the evaluation of the catalytic activity of Ar-BINMOLs and chiral diol analogues owing to moderate enantioselectivity, the present work and new [1,2]-Wittig rearrangement chemistry provides the basis for further synthetic applications involving Ar-BINMOLs as efficient chiral ligands to produce excellent chirality transfer in metal-catalyzed organic transformations. Finally, we found the chiral Ar-BINMOL ligand **2h** mediated titanium-promoted 1,2-addition of MeMgBr to aldehydes to give the desired products in good yields with excellent enantioselectivities (up to 92% *ee*). Further studies in our laboratory aimed at the application of the chiral Ar-BINMOL ligands described herein in chiral synthesis and asymmetric catalysis are currently underway.

Experimental Section

General: Flash column chromatography was performed over silica (200–300 mesh). ^1H and ^{13}C NMR spectra were recorded at 400 and 100 MHz, respectively, with an Advance (Bruker) 400 MHz NMR Spectrometer and were referenced to the internal solvent signals. Thin layer chromatography was performed with silica gel F254 TLC plates and visualized with ultraviolet light. HPLC was carried out with a Waters 2695 Millennium system equipped with a photodiode array detector. EI and CI mass spectra were performed with a Trace DSQ GC/MS spectrometer. Data are reported in the form of (*m/z*). The substrates including isobutyllithium were commercial available and used directly. Diarylmethanols and some of the [1,2]-Wittig rearrangement products were known and confirmed by the usual spectroscopic methods (^1H NMR, ^{13}C NMR, IR). The ESI-MS analysis of the samples was performed with an LCQ advantage mass spectrometer (ThermoFisher Company, USA), equipped with an ESI ion source in the positive ionization mode, and data was acquired with the Xcalibur software (Version 1.4). DCM was dried and distilled from CaH_2 . THF and Et_2O were distilled from sodium benzophenone. Aldehydes and titanium tetraisopropoxide were used after distillation. X-ray diffraction: data sets were collected with Bruker APEX DUO and Bruker APEX-II CCD diffractometers. Programs used: data collection Bruker APEX2,^[18a] data reduction Bruker SAINT, absorption correction for multi-scan, structure solution SHELX-97,^[18b] structure refinement SHELXL-97,^[18b] graphics Bruker SHELXTL.^[18b]

General Procedure for the Synthesis of Chiral Ar-BINMOL Derivatives 2a–2h by [1,2]-Wittig Rearrangement: The [1,2]-Wittig rearrangements were performed by stirring a solution of compound **1** (5 mmol) in anhydrous THF (30 mL) at -78°C , followed by the addition of a solution of *i*BuLi in hexane (1.23 M, 2.5 equiv.) under N_2 . The mixture was warmed slowly to room temperature, and then

the mixture was stirred at room temperature for one hour. After the reaction was completed, it was quenched with water and extracted with EtOAc. The combined organic layers were dried with Na_2SO_4 and concentrated in vacuo. The crude products were purified by column chromatography (silica gel, petroleum ether/EtOAc = 4:1) to afford the pure Ar-BINMOL **2**. Except for **2g**, the Ar-BINMOL products **2a–2f** and **2h** are known compounds and have been reported in our previous paper.^[11,12a] The syntheses of other chiral diols are presented in the Supporting Information.

(*R*)-2'-[(*S*)-(1,1'-Biphenyl-2-yl)(hydroxy)methyl]-[1,1'-binaphthalen]-2-ol (2g**):** ^1H NMR (400 MHz, CDCl_3): δ = 8.03 (d, J = 8.8 Hz, 1 H), 7.97 (d, J = 8.0 Hz, 1 H), 7.90 (d, J = 8.8 Hz, 1 H), 7.82–7.77 (m, 2 H), 7.50 (t, J = 7.2 Hz, 1 H), 7.38–7.34 (m, 1 H), 7.25–7.15 (m, 5 H), 7.08 (t, J = 8.0 Hz, 2 H), 7.02 (d, J = 8.8 Hz, 2 H), 6.91–6.89 (m, 1 H), 6.83–6.79 (m, 1 H), 6.58 (s, J = 6.8 Hz, 2 H), 6.24 (d, J = 8.4 Hz, 1 H), 5.99 (s, 1 H), 5.84 (s, 1 H), 3.10 (s, 1 H) ppm. ^{13}C NMR (100 MHz, CDCl_3): δ = 150.7, 141.9, 141.1, 140.0, 138.9, 133.5, 133.4, 130.2, 130.0, 129.1, 128.9, 128.8, 128.2, 128.1, 127.8, 127.5, 127.3, 126.8, 126.7, 126.3, 126.2, 126.1, 125.2, 125.0, 123.1, 117.8, 117.0, 70.7 ppm. IR (KBr): $\tilde{\nu}$ = 3540, 3057, 2922, 1620, 1595, 1512, 1478, 1452, 1341, 1278, 1195, 1144, 1004, 940, 840, 760, 741, 704, 635 cm^{-1} . MS (ESI $^-$): *m/z* = 902 [$2\text{M} - 2\text{H}$] $^+$. HRMS (EI): calcd. for $\text{C}_{33}\text{H}_{24}\text{O}_2$ [$\text{M} - 1$] $^+$ 451.1701; found 451.1776. $[\alpha]_{\text{D}}^{26}$ = +34.26 (c = 0.33, CHCl_3).

General Procedure for the Catalytic Asymmetric Addition of Grignard Reagents to Aldehydes: In a flame dried Schlenk tube, (*S,R*)-**2h** (21.3 mg, 0.05 mmol) and $\text{Ti}(\text{O}i\text{Pr})_4$ (0.295 mL, 1.0 mmol) were mixed in DCM (0.5 mL) at room temperature. After the resulting mixture was stirred for 5 min, 2-MeOC₆H₄CHO (0.5 mmol) in DCM (0.5 mL) was added. After 30 min, the mixture was cool to -20°C . A mixture of PhMgBr (1 mL, 1 M in THF) and $\text{Ti}(\text{O}i\text{Pr})_4$ (0.295 mL, 1.0 mmol) in DCM (3 mL) stirred for 30 min at room temperature was added dropwise to the mixture over a period of 5–10 min. The reaction was stirred at -20°C for 10 h. After the reaction was completed as monitored by TLC, the reaction mixture was quenched with 1 N aqueous HCl and extracted with DCM (3 $\times$ 10 mL). The combined organic layers were washed with brine, dried with Na_2SO_4 , filtered, and the solvent was removed under reduced pressure. The crude product was purified by column chromatography on silica gel (petroleum ether/ethyl acetate, 10:1) to give the pure product (76.0 mg, 71% yield, 71% *ee*).

CCDC-893055 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

Supporting Information (see footnote on the first page of this article): General remarks, NMR and IR spectroscopic data for the products.

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